

QoE Enhancement of SVC Video Streaming over Vehicular Networks Using Cooperative LTE/802.11p Communications

Elias Yaacoub, Fethi Filali, and Adnan Abu-Dayya

Abstract—Real-time streaming of scalable video coded (SVC) videos in vehicular networks is investigated, and novel cooperative vehicle-to-vehicle (V2V) communication methods are proposed. The proposed techniques are based on grouping the moving vehicles into cooperative clusters. Within each cluster, the long term evolution (LTE) system is used to send the data over long range cellular links to a selected cluster head, which multicasts the received video over IEEE 802.11p to vehicles in its cluster. Error concealment techniques are used to compensate the loss of frames that are not delivered on time for real-time video streaming. In addition, resource allocation to select the best subcarriers for LTE transmission is performed in order to enhance the received video quality. Moreover, metrics related to both quality of service (QoS) and quality of experience (QoE) are studied and analyzed for various video sequences of different characteristics. The V2V video streaming techniques are extended to the case of vehicle-to-infrastructure (V2I) communications. Simulation results show that the proposed methods lead to enhanced QoS and QoE compared to the non collaborative scenarios, and their performance tradeoffs compared to recent methods from the literature are discussed.

Index Terms—Video streaming, scalable video coding, quality of experience, quality of service, vehicle-to-vehicle communications, vehicle-to-infrastructure communications, LTE, IEEE 802.11p.

I. INTRODUCTION

Intelligent transportation systems (ITS) are receiving significant research attention. The main objective of ITS is to reach enhanced safety on the roads by using vehicle-to-vehicle (V2V) communications and vehicle-to-infrastructure (V2I) communications [1]. In addition to the primary goal of road safety, V2V and V2I are also used for traffic efficiency [2] and for the delivery of infotainment services, which is the application considered in this paper.

The fast varying channel conditions and connectivity in vehicular networks represent additional challenges compared to wireless communications in low mobility scenarios [3]. In fact, wireless communications in ITS systems should operate with minimal errors while reliably delivering vital data in real time. Hence, challenges imposed by the dynamic surroundings such as multipath fading, shadowing, and path loss should be surmounted by the ITS system [4].

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Therefore, in this paper, collaborative V2V communication methods are proposed to enhance the quality of the video transmission, measured in terms of the peak signal to noise ratio (PSNR) and of a quality of experience (QoE) metric derived in [5] and extended in this paper to the investigated system model. The proposed methods are based on grouping vehicles into collaborative clusters where the cluster members cooperate to share the video transmitted by a cellular base station (BS). We consider scalable video coded (SVC) sequences where the video is encoded into multiple layers: a base layer providing a basic quality level and one or more enhancement layers that refine the video quality [6]. Hence, the granularity of quality variation in SVC allows video transmission to adapt to the wireless channels in vehicular networks and their fast variations. In addition, the proposed methods are extended to the scenario of V2I communications.

Clustering in vehicular networks is investigated in several papers, e.g. [7–9]. In [7], clustering is performed in vehicular ad-hoc networks (VANETs) by dividing the service area into several region units in order to associate each of these units with a non-overlapping radio channel pool. The objective is to limit the number of vehicles in each region unit in order to reduce the contentions for radio channels and increase throughput. However, clustering in [7] consists of defining the regions containing vehicle groups or clusters, without selecting a specific cluster head (CH) to handle communications between cluster members: the members still communicate independently with each other, using the radio channels dedicated to their “cluster”. The CH selection is handled in [8], where two CHs are selected for each group of vehicles: one at the front and another at the rear. The aim is to reduce the load on the single CH when all the traffic is routed through that CH, by providing another CH thus helping to enhance quality of service (QoS) and reduce energy consumption. The CH selection in [8] is based on the vehicle position. In [9], a clustering method based on both vehicle position and mobility is proposed. The algorithm is named CPM (center position and mobility), and it is shown to perform well compared to other clustering methods. Although the results in [9] are presented for the single cluster case, a method to extend the approach to a multiple cluster scenario is presented. It consists of dividing the area into several regions, and implementing the CPM algorithm to select a CH for each region.

The methods in [7–9] assume the vehicles know the positions of each other, which necessitates the exchange of position information, like GPS. They are also mostly focused

on safety applications, where a vehicle communicates with its neighbors to relay accident information for example. Compared to previous work, this paper presents an approach that does not require the exchange of position information between vehicles. It relies on simple physical layer techniques used by wireless communication standards to determine the quality of communication links between the vehicles on one hand, and between the vehicles and the cellular BSs on the other hand, in order to form the clusters and select the CHs. Moreover, the work in this paper mainly targets infotainment scenarios with video streaming applications.

Hence, this paper combines clustering for V2V communications and SVC video transmission. In [10], video streaming is studied with single layer videos (not SVC) in low mobility non-vehicular scenarios. In [11], collaborative vehicular communications for data transmission are studied within a single cluster, assumed already formed, without proposing any clustering techniques and without treating video streaming. SVC video streaming over wireless networks has been considered in [12], and was investigated in [13] for a multiple-input multiple-output scenario. However, SVC transmission over collaborative vehicular networks with dynamic vehicle clustering and cellular resource allocation has not been adequately treated in the literature, to the best of the authors' knowledge.

The main contributions of this paper can be summarized as follows:

- Customizing the QoE metric of [5] to accommodate SVC video transmission and losses over wireless channels.
- Investigating different methods to enhance QoS and QoE performance during the transmission of the layers of SVC coded videos in V2V and V2I communications.
- Proposing resource allocation algorithms for the joint operation of short range (SR) V2V communications and long range (LR) LTE communications in vehicular networks.
- Simulating the presented methods using LTE and 802.11p taking into account wireless resource allocation, correlation in log-normal shadowing, and modulation and coding schemes (MCS) adopted by the LTE [14, 15] and IEEE 802.11p [16, 17] standards.

The rest of the paper is organized as follows. The system model is presented in Section II. In Section III, SVC video coding and the error concealment technique used in this paper are discussed, along with the QoS/QoE metrics used to assess the streaming quality. Sections IV and V describe proposed collaborative V2V communications techniques and their V2I extensions, respectively. Simulation results are presented and discussed in Section VI. Finally, conclusions are drawn in Section VII.

II. SYSTEM MODEL

The system model is presented in Fig. 1. The investigated scenario consists of a number K of vehicles moving on a highway stretch. The vehicles are interested in downloading a video sequence from a remote server through the BSs of the LTE network covering the road. The content is assumed of

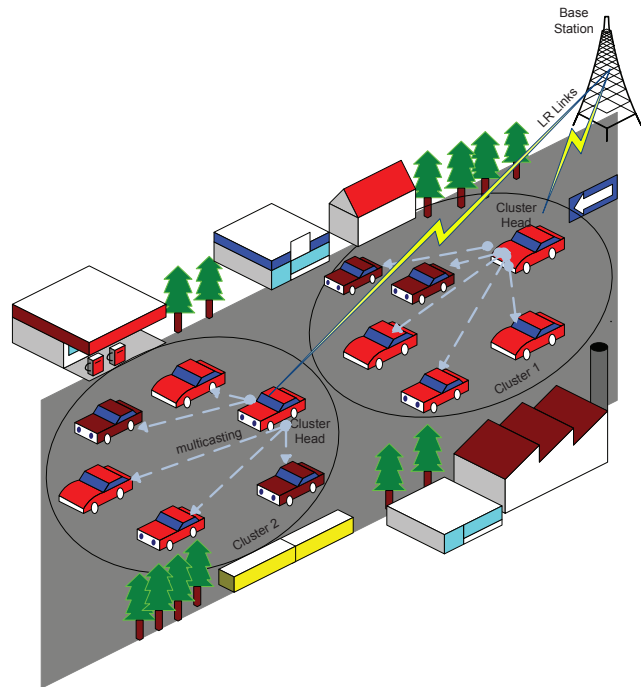


Fig. 1. System model.

common interest to the vehicles, e.g., live broadcast service during the World Cup or other sports events, advertisement videos for facilities in the surrounding area (e.g., malls, shopping centers, tourist information), etc. Since the vehicles are moving fast, this poses a challenge on the LTE network to distribute the video at acceptable QoS/QoE. Hence, neighboring vehicles can form cooperative groups or clusters to distribute the video file: a CH receives a video layer from the BS over LR LTE links, and multicasts it to the vehicles in its cluster using SR V2V communications over IEEE 802.11p.

LTE communication is considered on the LR links. In LTE, the available spectrum is divided into resource blocks (RB) consisting of 12 adjacent subcarriers, allocated in a 0.5 ms time slot. The shortest assignment unit consists of two consecutive slots, i.e., for a duration of 1 ms, which is the duration of one transmission time interval (TTI) [14, 15]. The SR V2V links use the WLAN 802.11p protocol which is different from the 802.11 standard in that it added support for WLAN in a vehicular environment [16, 17]. It operates in the 5.8 GHz and 5.9 GHz frequency bands.

A. Parameters and Variables

We denote by $\mathcal{R}$ the set of CHs acting as vehicular relays, and by $\mathcal{C}_k$ the set of vehicles in the cluster served by relay k , with $|\mathcal{C}_k| = C_k$, where $|\cdot|$ represents set cardinality. In addition, we denote by R_{0d} the transmission rate on the LR links from the BS to destination d (could be a CH in the collaborative V2V approach or any vehicle in the traditional approach), and by R_{kj} the achievable transmission rate on the SR links from vehicle k to vehicle j .

B. Channel Model

The channels are modeled by pathloss, shadowing and fading. The channel gain on the link between source s and destination d over subcarrier x is given by:

$$H_{s,d}^{(x)} = (-\kappa - v \log_{10} l_{s,d}) - \xi_{s,d} + 10 \log_{10} F_{s,d}^{(x)}. \quad (1)$$

In the presence of relays, the source is either the BS transmitting to the relays (destination), or it could be the relay vehicle transmitting to the other vehicles. In the traditional case without relays, the source is the BS and the destinations are the vehicles.

In (1), the first factor captures propagation loss, with $l_{s,d}$ the distance between s and d , κ the pathloss constant, and v the path loss exponent. The second factor, $\xi_{s,d}$, captures log-normal shadowing with a standard deviation σ_ξ , whereas the last factor, $F_{s,d}^{(x)}$, corresponds to fast Rayleigh fading power with a Rayleigh parameter a selected such that $E\{|a|^2\} = 1$.

In the channel model, spatial shadowing correlation is taken into account since shadow fading values depend on the fixed location of obstacles [18]. Spatial correlation can be described as a measure of how fast the local mean power evolves as the vehicle moves along a certain route [19]. We will apply the correlated shadowing model of [19] and [20] where the shadowing correlation is expressed as:

$$\Lambda_\xi(\Delta l_s, \Delta l_d) = \exp\left(-\frac{\Delta l_s + \Delta l_d}{l_{cor}} \ln 2\right), \quad (2)$$

where Δl_s and Δl_d represent the movements of the transmitter (source) and receiver (destination), respectively, from positions l_s and l_d to positions $l_s + \Delta l_s$ and $l_d + \Delta l_d$. The parameter l_{cor} is the decorrelation distance taken to be 20 meters for urban environment. As the vehicles move, the model of (2) can be applied to determine the correlation between shadowing values at the different vehicle positions, in addition to the correlation between shadowing values of the same moving vehicle on the link with the BS (by setting $\Delta l_s = 0$ since the BS is fixed). Consequently, (2) can be used to determine the shadowing values as follows:

$$\begin{aligned} \xi_{s,d}(l_s + \Delta l_s, l_d + \Delta l_d) &= \xi_{s,d}(l_s, l_d) \cdot \Lambda_\xi(\Delta l_s, \Delta l_d) \\ &+ (1 - \Lambda_\xi(\Delta l_s, \Delta l_d)) \cdot \xi_{s,d}^{(new)}. \end{aligned} \quad (3)$$

In (3), the shadowing value at positions $(l_s + \Delta l_s, l_d + \Delta l_d)$ has a component equal to the shadowing value at positions (l_s, l_d) , with the amount of similarity determined by $\Lambda_\xi(\Delta l_s, \Delta l_d)$ (first term on the right hand side of (3)), in addition to an independent shadowing component corresponding to the new location $\xi_{s,d}^{(new)}$ (second term on the right hand side of (3)).

For fast Rayleigh fading, a block fading model is considered, where the fast fading remains constant for a fixed time T_{dec} which is the channel de-correlation time. Then the channel conditions change and remain constant for another T_{dec} , and so on.

TABLE I
DISCRETE RATES AND SNR THRESHOLDS WITH THE MODULATION AND CODING SCHEMES USED FOR LONG RANGE COMMUNICATIONS

Modulation	Coding Rate	r_t (bits/symb)	η_t (dB)
No Transmission	0	0	$-\infty$
QPSK	78/1024	0.1523	-7.2
QPSK	120/1024	0.2344	-5.8
QPSK	193/1024	0.3770	-3.7
QPSK	308/1024	0.6016	-1.5
QPSK	449/1024	0.8770	0.1
QPSK	602/1024	1.1758	1.8
16-QAM	378/1024	1.4766	4.1
16-QAM	490/1024	1.9141	6.8
16-QAM	616/1024	2.4063	8.9
64-QAM	466/1024	2.7305	10.7
64-QAM	567/1024	3.3223	11.6
64-QAM	666/1024	3.9023	13.9
64-QAM	772/1024	4.5234	16.2
64-QAM	873/1024	5.1152	18.1
64-QAM	948/1024	5.5547	22.4
64-QAM	1 (uncoded)	6.0	26.8

C. Data Rates

Given the transmit power $P_{s,d}^{(x)}$ that source s is using in order to transmit to destination d over subcarrier x , the channel gain $H_{s,d}^{(x)}$ of the link between s and d over subcarrier x , the antenna gain of the source G_s , the antenna gain of the destination G_d , and the thermal noise power σ^2 , the received signal-to-noise ratio (SNR) $\gamma_{s,d}^{(x)}$ on the link between s and d over subcarrier x can be calculated as:

$$\gamma_{s,d}^{(x)} = \frac{P_{s,d}^{(x)} G_s G_d H_{s,d}^{(x)}}{\sigma^2}. \quad (4)$$

The actual data rate $r_{s,d}^{(x)}$ is an increasing function of the SNR: $r_{s,d}^{(x)} = \mathcal{F}(\gamma_{s,d}^{(x)})$. Continuous rates can take any non-negative real value according to the Shannon capacity formula $\log(1 + \gamma_{s,d}^{(x)})$. Conversely to continuous rates, discrete rates represent the quantized bit rates r_n achievable in a practical system as follows:

$$r_{s,d}^{(x)}(\gamma_{s,d}^{(x)}) = \begin{cases} r_0, & \eta_0 \leq \gamma_{s,d}^{(x)} < \eta_1 \\ r_1, & \eta_1 \leq \gamma_{s,d}^{(x)} < \eta_2 \\ r_2, & \eta_2 \leq \gamma_{s,d}^{(x)} < \eta_3 \\ \vdots & \vdots \\ r_{L-1}, & \eta_{L-1} \leq \gamma_{s,d}^{(x)} < \eta_L \end{cases} \quad (5)$$

where η_n represents the SNR target in order to achieve the rate r_n with a predefined bit error rate (BER). Note that in the limit, we have $r_0 = 0$, $\eta_0 = -\infty$ (in dB), and $\eta_L = \infty$.

In this paper, we consider discrete sets of Modulation and Coding Schemes (MCS) for the rate calculations according to (5), as is the case in practical standards. Thus, we adopt the discrete rates used in LTE and 802.11p. The MCS used in LTE can be found in [15]. They are listed in Table I along with the SNR targets used in this paper for each MCS (estimated based on Figure A.4 in [21]). The MCS used in 802.11p in addition to their corresponding data rates can be found in [16, 17] and are shown in Table II.

TABLE II
THE 802.11P DISCRETE RATES AND SNR THRESHOLDS FOR DIFFERENT
MODULATION AND CODING RATES

Modulation	Coding Rate	Rate(Mbps)	η (dB)
BPSK	1/2	3	-2.2
BPSK	3/4	4.5	-0.3
QPSK	1/2	6	1.3
QPSK	3/4	9	4.6
16QAM	1/2	12	7
16QAM	3/4	18	11.15
64QAM	2/3	24	14
64QAM	3/4	27	16

The long range discrete rate R_{0d} of a destination d , communicating with the BS at a rate $r_{s,d}^{(x)} = r_n$ bits/symbol over the subcarriers of the RBs allocated to it, can be calculated as follows:

$$R_{0d} = \frac{r_n \cdot N_{RB}^{(d)} \cdot N_{SC}^{RB} \cdot N_{Symb}^{SC} \cdot N_{Slot}^{TTI}}{T_{TTI}}, \quad (6)$$

where $N_{RB}^{(d)}$ is the number of RBs allocated to d , N_{SC}^{RB} is the number of subcarriers per RB (equal to 12 in LTE), N_{Symb}^{SC} is the number of symbols per subcarrier during one time slot (set to six or seven in LTE, depending whether an extended cyclic prefix is used or not), N_{Slot}^{TTI} is the number of time slots per TTI (two 0.5ms time slots per TTI in LTE), and T_{TTI} is the duration of one TTI (1ms in LTE) [14].

III. VIDEO ENCODING AND ERROR CONCEALMENT

Each SVC layer is encoded into groups of pictures (GOPs), with each GOP consisting of one I-frame and a sequence of P-frames and having a duration T_{GOP} . When a GOP becomes available at the BS, coming from a video streaming server, it should be sent to the moving vehicle within a duration of T_{GOP} . When T_{GOP} has elapsed, all the frames that are not received by the vehicle are assumed lost, and the transmission of a new GOP begins. Due to the interdependencies of the video frames, the loss of a frame in a GOP is considered to lead to the loss of all subsequent frames in the GOP, until the next I-frame is received. The loss of an I-frame leads to the loss of all the frames in the GOP [22].

With SVC, the base layer provides a basic level of quality and can be decoded independently of the enhancement layers, whereas the enhancement layers serve to refine the base-layer quality. An enhancement layer cannot be used without the base layer, which is the most critical part of the scalable representation. In this paper, we use SVC with two layers: a base layer L_0 in QCIF format at coarse resolution, and an enhancement layer L_1 in CIF format at higher resolution. Both layers are considered to have the same frame rate. We use video encoded in H.264 format using the joint scalable video model (JSVM) software. Within each layer, a GOP of N_G frames is considered to consist of one I-frame and $N_G - 1$ P-frames in this paper.

A. Video Error Concealment

When video frames are lost, error concealment techniques are used. The most common approach for error concealment is

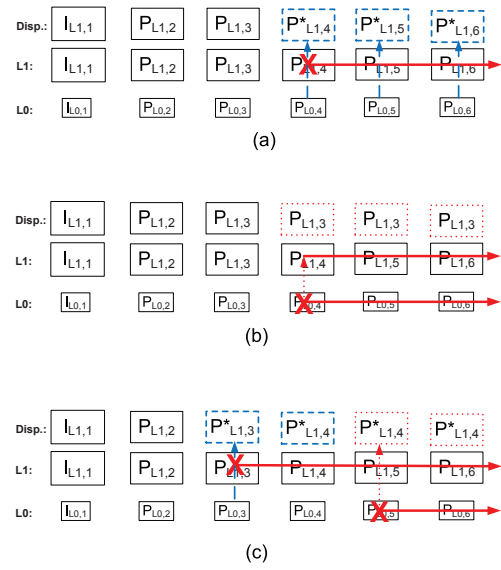


Fig. 2. Error concealment example with SVC: (a) L_1 frame lost; (b) L_0 frame lost; (c) L_1 frame lost first then L_0 frame lost. The superscript '*' indicates the interpolation of a frame from the dimensions of L_0 to the dimensions of L_1 .

known as the previous frame concealment method. It consists of repeating the last correctly received frame until the next I-frame is received [22]. Although more advanced techniques can be used to reduce the error, the purpose of this paper is to study the performance of the proposed video transmission techniques in vehicular communications. Hence, comparing different concealment techniques is beyond the scope of this work.

An error concealment example is shown in Fig. 2 for the case of SVC with two layers. When a frame from layer L_1 is lost first without losses from L_0 (Fig. 2 (a)), the corresponding frame from layer L_0 is interpolated to the L_1 resolution, until the next L_1 I-frame is received. When a frame from layer L_0 is lost first without losses from L_1 (Fig. 2 (b)), the corresponding frame in L_1 is lost. Hence, the last correctly received frame at the L_1 resolution is used for error concealment. When a frame from layer L_1 is lost first followed by a loss from L_0 (Fig. 2 (c)), then the last interpolated frame at the L_1 resolution is used for error concealment, until the next I-frame is received. When the opposite occurs, i.e., a frame from layer L_0 is lost first followed by a loss from L_1 , the scenario of Fig. 2 (b) applies, since a loss from the base layer leads to the loss of the enhancement layers.

B. Video QoS/QoE Metrics

PSNR is generally used to measure QoS in video transmission, e.g. as in [22], and it is calculated from the mean-squared error distortion. It should be noted that a differentiation should be made between two types of distortion: source distortion, that depends on the compression method at the source, and loss distortion, caused by lost frames during transmission over the wireless channels. In this paper, videos already encoded are considered (and hence techniques to lower source distortion are beyond the scope of this paper). Total distortion is calculated between the original video sequence and the received sequence, obtained after transmission over

the wireless channels and the implementation of the error concealment approach described in Section III-A and Fig. 2. The distortion for replacing a frame f , of dimensions $N_1 \times N_2$ pixels, with an estimated frame $\hat{f}$ is given by [22]:

$$D(f, \hat{f}) = \frac{1}{N_1 N_2} \sum_{n_1=0}^{N_1-1} \sum_{n_2=0}^{N_2-1} \left(f(n_1, n_2) - \hat{f}(n_1, n_2) \right)^2 \quad (7)$$

where $f(n_1, n_2)$ indicates the pixel value of frame f at position (n_1, n_2) . The PSNR in this case is given by:

$$P_{\text{SNR}}(f, \hat{f}) = \frac{(2^B - 1)^2}{D(f, \hat{f})} \quad (8)$$

with B the number of bits used to encode a single pixel in the picture frame.

On the other hand, QoE is gaining significant interest as a method to quantify the multimedia experience of mobile users, e.g., see [5, 23, 24]. A survey of QoE techniques is presented in [24]. QoE tries to measure the QoS as it is finally perceived by the end-user. For example, following [25], the QoE can be related to the PSNR as follows:

$$Q = \frac{1}{1 + e^{b_1(P_{\text{SNR}} - b_2)}} \quad (9)$$

where b_1 and b_2 are parameters that depend on the video characteristics, and P_{SNR} is the PSNR expressed in dB. In (9), $Q = 0$ indicates the best quality and $Q = 1$ indicates the worst quality. In [23], an empirical QoE metric taking into account PSNR, spatial resolution, frame rate, in addition to spatial and temporal variances was derived. A QoE metric derived in [5], based on the metric in [25], is expressed as:

$$Q_m = Q_{\text{max}} \left(1 - \frac{1}{1 + e^{b_1(P_{\text{SNR}} - b_2)}} \right) \cdot \frac{1 - e^{-b_3 \frac{f_R}{f_{R, \text{max}}}}}{1 - e^{-b_3}} \quad (10)$$

where b_1 , b_2 , and b_3 are parameters that depend on the video characteristics, Q_{max} is a constant corresponding to maximum quality, f_R is the frame rate at which the video is displayed, and $f_{R, \text{max}}$ is the maximum frame rate. The QoE derivations take into account user experience while playing the video and are not inherently designed to assess the transmission over wireless channels. Using the approach of Section III-A depicted in Fig. 2, the same frame rate can be maintained after error concealment (i.e. $f_R = f_{R, \text{max}}$), and hence (10) can be simplified to:

$$Q_m = Q_{\text{max}} \left(1 - \frac{1}{1 + e^{b_1(P_{\text{SNR}} - b_2)}} \right) \quad (11)$$

Hence, in this paper, we use (11) with $Q_{\text{max}} = 100$, thus displaying QoE on a scale from 0 to 100.

In practice, during the streaming of a stored video at the streaming server or BS, the video characteristics can be extracted offline and used to determine b_1 , b_2 , and b_3 in the QoE metrics. However, these parameters are difficult to extract during the streaming of a live video (e.g. live broadcast of a sports match). Although an approach for their dynamic real-time estimation is presented in [5], the proposed methods in Sections IV and V are independent from these parameters. They tend to maximize the performance by transmitting the

video layers in parallel, with the frames transmitted in decreasing order of priority. The QoE metric is then used to assess the performance after implementing the proposed techniques.

IV. PROPOSED V2V VIDEO STREAMING METHOD

In this section, the proposed approach is presented. It is based on dynamic cluster formation and on LTE resource allocation.

A. Cluster Formation Method

Starting with the vehicles having worst LR channel conditions with the BS, we find for each vehicle j the parent p_j from which it can receive the video at the highest communication rate, or, equivalently, the shortest time. If j is a cluster head, all members of C_j are moved to C_{p_j} if the distribution of the video from p_j to j and all the members of C_j is faster than having an independent cluster C_j with j as cluster head. The details of the proposed approach are presented below:

- **Step 1:** Sort the vehicles in increasing order of achievable data rates on the LR LTE links in the absence of cooperation. After this step, vehicles $j = 1$ and $j = K$ would be the vehicles having the worst and best channel conditions on the LR, respectively. In this step all vehicles are still connected to the BS; hence we set $p_j = 0$ for all j .
- **Step 2:** Start from vehicle $j = 1$.
- **Step 3:** For vehicle j , find the parent node (could be another vehicle or the BS) p_j from which j and all the members of C_j (if there are any vehicles other than j) can communicate at the highest data rate, i.e.:

$$p_j = \arg \max_{k; k > j} \left(\min_{i \in C_j} R_{ki} \right) \quad (12)$$

The minimum rate is used in (12) since the CH multicasts the data to its cluster members, i.e. transmits it once to all members. Hence, the minimum rate is used to guarantee successful communication with all the vehicles in the cluster.

- **Step 4:** Break the connection of j with the BS, and the connection of all other members of C_j with j , and set p_j as the direct parent of j and all vehicles in C_j if:

$$\frac{1}{R_{0p_j}} + \frac{1}{\min_{i \in (C_{p_j} \cup C_j), i \neq p_j} R_{p_j i}} < \frac{1}{R_{0j}} + \frac{1}{\min_{i \in C_j, i \neq j} R_{ji}} \quad (13)$$

i.e., move all members of C_j to C_{p_j} if the video is distributed faster than having an independent cluster with j as cluster head receiving the data from the BS. In fact, on both sides of the inequality in (13), the first term corresponds to the time needed to transmit 1 bit from the BS to the CH (whether p_j or j), and the second term represents the time to transmit that bit from the CH to the members of its cluster. If the inequality is verified, the clusters are merged: $C_{p_j}^{(\text{new})} = C_{p_j}^{(\text{old})} \cup C_j$. Otherwise, they are kept separate.

- **Step 5:** Increment j and repeat Steps 3-5 on the vehicles whose order is greater than j in the sorted list.

This clustering method can revert back smoothly to non-cooperative LTE communications with the BS when the user density is low or when the cars are too close to the BS. In fact, when the communication between a vehicle and the BS can take place at a higher rate than with any of the neighboring vehicles, then that vehicle will be connected directly to the BS and will not be part of a cluster (or, equivalently, will form a cluster of its own consisting of a single vehicle communicating directly with the BS without having other cluster members). This scenario corresponds to having $p_j = 0$ in (12).

B. LTE Resource Allocation

In this section, we present the LTE resource allocation techniques adopted on the LR links. We denote by $r_{k,z}$ the rate achievable by vehicle k over RB z at a given channel realization. To implement the proposed clustering approach of Section IV-A, we need to determine the individual LR rates of the vehicles, in order to use them in (13). Hence, we find the pair $(k^*, z^*) = \{\arg \max_{k,z} r_{k,z}\}$, and we set $R_{0k^*} = r_{k^*,z^*}$. Then the same approach is repeated for the other vehicles and RBs (after excluding k^* and z^*).

To perform LR multicasting after implementing the approach of Section IV-A, we dedicate a single RB in the cell to multicast the video stream to the vehicles. In the case of multicasting, transmission on a given RB is limited by the rate achieved by the vehicle having the worst channel conditions on that RB. Thus, the BS selects the RB having the highest minimum rate, i.e., according to the following:

$$z^* = \left\{ \arg \max_z \left(\min_{k; p_k=0} r_{k,z} \right) \right\} \quad (14)$$

In the non-cooperative scenario, all vehicles are involved in the allocation of (14). Thus, the achievable multicasting rate on the LR is affected by the vehicle having the worst channel conditions. However, in the collaborative approach, clustering is performed according to the method described in Section IV-A. Afterwards, only the cluster heads are involved in the minimization in (14). Since cluster heads are selected as such because they have the highest achievable LR rates, then the outcome of (14) is expected to lead to significantly higher LR multicasting rates, since multicasting is performed on a subset of the participating vehicles, with the members of the subset having better LR channel conditions (and hence higher rates) than the others.

C. Cluster Formation for SVC Transmission

The approach of Sections IV-A and IV-B corresponds to a single layer video. With SVC, the proposed approach consists of transmitting the layers simultaneously, with each layer sent on a different channel to ensure faster distribution and better QoS/QoE. Hence, in this section, we extend the previous approach to SVC with a base layer and one enhancement layer.

For SVC clustering, the clustering approach of Section IV-A is implemented for L_0 . Another set of CHs is needed for the same cluster in order to transmit L_1 simultaneously with the CHs transmitting L_0 , without causing any interference or collisions between the two sets of transmitting CHs. Therefore,

for L_1 , the CHs of L_0 are given fictitious rates on the LR, i.e. we set $R_{0k} = \epsilon \ll 1$ (we use ϵ to avoid division by zero in the equations), then the same approach of Section IV-A is implemented for L_1 . With $R_{0k} = \epsilon$, the CHs of L_0 need a very long time to receive L_1 on the LR. Thus, the outcome of the approach will be a different set of CHs for L_1 , and the CHs of L_0 will be cluster members receiving L_1 on the SR. This method leads to having the best set of CHs to transmit L_0 and the second best set to transmit L_1 on the SR. Hence, for example, assuming that for a given cluster of cars, CH A is selected as the best CH to transmit L_0 , the proposed approach will lead to the selection of another CH B within the same cluster for the transmission of L_1 . In this case, during the transmission of L_1 , CH A will be a cluster member receiving the data from CH B, similarly to the other cluster members. To ensure the simultaneous transmission of the two layers, different channels should be selected for each layer to avoid interference on both the LR LTE links and the SR V2V links.

Therefore, after determining the cluster heads for each layer, one RB is selected for multicasting simultaneously each of the layers on the LR. Thus, RB z_1^* is obtained after applying (14) to the CHs of L_0 , and RB $z_2^* \neq z_1^*$ is obtained after applying (14) to the CHs of L_1 while excluding z_1^* from the search (since it is already allocated for L_0). In the non-collaborative case, z_1^* is used to transmit L_0 after implementing (14) on all vehicles, and z_2^* is used to transmit L_1 after implementing (14) on all vehicles while excluding z_1^* from the search.

Although resource allocation is performed for LR LTE links, fixed channel assignments are assumed on the SR V2V links. In fact, LTE resource allocation is performed in a centralized way by the BS. On the other hand, in IEEE 802.11p, the cars have to sense the channel and perform a contention approach for random access based on collision avoidance. There are seven orthogonal channels in IEEE 802.11p, including one control channel (channel no. 178) and two reserved channels (channels no. 178 and 184), used mainly for safety applications [26]. This leaves only four channels to be used. With two CHs per cluster in the SVC case, the four channels can serve only two neighboring clusters without causing interference or collisions. Therefore, the best solution would be to use channels 174 and 180 for one cluster, and channels 176 and 182 for another neighboring cluster, if any. Then, these pairs of frequencies can be reused for clusters further away. It would be safe to ignore interference in this case since a sufficient distance would be separating any two clusters using the same frequency. For example, in the scenario of Fig. 1, Cluster 1 would use channels 174 and 180, whereas Cluster 2 would use channels 176 and 182. If an additional cluster, denoted by Cluster 3, is positioned after Cluster 2, it would use the same channels as Cluster 1. Thus, it will not interfere with neighboring Cluster 2. Moreover, given the lower transmit power for SR V2V communications (compared to BS transmit power for LR LTE transmissions) and the distance separating it from Cluster 1, the interference power at Cluster 1 would be limited and thus will be ignored in this paper.

D. Partial Collaboration Approach

Another approach consists of partial collaboration, i.e. one CH is determined per cluster according to the method of Section IV-A, and used for the transmission of the base layer. The enhancement layer is in this case multicast by the BS to all vehicles using the RB leading to the highest multicasting rate. This allows the system to secure the transmission of the base layer using high data rates achievable over SR communications.

E. Implementation in Practical V2V Scenarios

Due to the fast channel variations in vehicular networks, the overhead due to repeating the clustering approach of Section IV-C each time the channel conditions change is non negligible. In practice, vehicles forming a cluster are generally moving relatively close to each other. Hence, we propose a practical implementation of the previous method by distinguishing between a clustering phase and a dynamic CH selection phase:

- Cluster formation phase: The approach of Section IV-C is used to determine the clusters and the members of each cluster. This clustering phase is repeated on a slow timescale, in the order of seconds, e.g. 5-10 seconds.
- Dynamic cluster head variation based on LTE resource allocation: Within each cluster, CHs are varied dynamically based on channel variations on the LR LTE links at a faster timescale. Hence, although the clustering phase leads to a preliminary CH for each cluster, this CH can be changed at the next TTI or T_{dec} when the state of the wireless channel varies, although the members of each cluster remain the same until the next cluster formation iteration.

In the simulations, the cluster formation phase is repeated every eight seconds, and the CH selection within each cluster is done every TTI as supported by LTE. Most of the focus in the V2V literature (e.g. [7–9]) is on road safety applications. In such scenarios, cars need to exchange real time information about the road status (e.g., accidents), and would not afford the delay of routing it through a cellular system to reach back neighboring cars a few meters away. However, in the scenario of this paper, the interest is in entertainment applications where cars are assumed to be connected to a cellular BS from which they will stream a video file (i.e. the initial source of information is a video server connected to the cellular network, not a vehicle on the road). Cellular coverage is easily available everywhere and state-of-the-art technologies can support communication at high speeds [27]. Thus, in case contact is lost between a CH and some of its cluster members due to high speed and mobility environments, while the implementation of the practical approach is performed at regular intervals, the cars can revert back to receiving the video directly from the BS on the LR LTE link. Nevertheless, in the simulations of Section VI-A in this paper, we consider a lower bound on this situation: We assume that the frames are lost (and perform error concealment as described in Section III) until the connection is established again with the CH or a new clustering has been performed. This way, BS resources

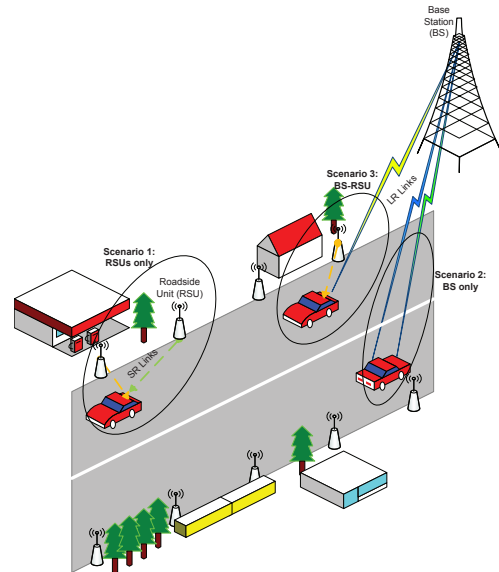


Fig. 3. Investigated V2I scenarios.

are saved by using one RB for each layer for all users through multicasting, instead of using a pair of RBs for each user with unicasting, which would drain quickly all the available RBs for streaming a single video and would not allow the BS to serve other users or applications. Thus, the approach takes advantage of the good LR channel conditions of certain users compared to others, by selecting them as CHs, and uses SR communications with the CHs to relay the data to other users over V2V links.

It should be noted that in the proposed approach, each cluster has two CHs, one for each video layer. However, the clusters themselves do not overlap; i.e. there is no intersection between a given cluster (having a pair of CHs) and another cluster (having another pair of CHs). This allows each car to receive a given video layer from a single source: either the BS or the CH of its cluster.

V. V2I VIDEO STREAMING METHODS

In this section, SVC video streaming methods over V2I communications are presented. In V2I systems, communications can take place between the vehicles and roadside units (RSUs) on the SR over IEEE 802.11p, or between the vehicles and cellular BSs over LR LTE cellular links. It is assumed that the BSs and RSUs are connected via some backbone network to the video servers where the videos can be retrieved. The methods of this section can be considered an extension of the V2V methods of Section IV, after replacing the CHs by fixed RSUs on the side of the road.

Three scenarios, shown in Fig. 3, are investigated in this paper:

- Scenario 1 - RSUs only: The video is transmitted to a moving vehicle via RSUs using SR communications over IEEE 802.11p. The RSU having the best channel conditions with the vehicle, and hence capable of transmitting at the highest data rate to that vehicle, is selected to transmit the base layer. The enhancement layers are transmitted over the second best RSU, i.e.



Fig. 4. Video sequences investigated in the simulations.

the RSU having the best channel conditions among the remaining RSUs (after excluding the RSU transmitting the base layer).

- Scenario 2 - BS only: The video is transmitted to a moving vehicle using LR LTE cellular links. The RB having the best channel conditions with the vehicle, and hence capable of supporting transmission at the highest data rate to that vehicle, is selected to transmit the base layer. The enhancement layers are transmitted over the second best RB, i.e. the RB having the best channel conditions among the remaining RBs (after excluding the RB transmitting the base layer).
- Scenario 3 - BS-RSU: This is a joint collaborative scenario between BSs and RSUs. The base layer is transmitted over the RSU having the best channel conditions with the vehicle using 802.11p, and the enhancement layers are transmitted over the RB having the best channel conditions with the vehicle using LTE.

VI. RESULTS AND DISCUSSION

The simulations are implemented in Matlab. A straight highway stretch with one travel direction and a length of 1 km, as in [8], is considered. the width of the highway is set to 50 m. An LTE BS is deployed on the side of this highway stretch, at a distance of 500 m from its origin. An LTE bandwidth of $W_{\text{tot}} = 5$ MHz, subdivided into $N_{\text{RB}} = 25$ RBs of 12 subcarriers each is considered. The BS total transmit power is set to 5 W, subdivided equally among RBs.

The following video sequences, with $N_G = 15$ frames and $T_{\text{GOP}} = 1$ sec, are used in the simulations: Foreman, Football, City, and Ice. The parameters (b_1, b_2) are set to (0.34, 29.09) for Foreman, (0.34, 25.9) for Football, (0.34, 26.3) for City, and (0.34, 31.24) for Ice, as in [5]. Snapshots of these video sequences are shown in Fig. 4. In each iteration, 120 frames from a given sequence, corresponding to eight seconds of video, have to be distributed to the vehicles. The results are averaged over 50 iterations for each video sequence. In each iteration, vehicles are placed at random positions on the highway. They are interested in streaming a video file on the move. Each vehicle is assumed to have an initial speed selected in the range between 70 and 100 km/hour, which corresponds to practical speeds on a highway. As the vehicle moves along the highway, its speed can vary by increasing or decreasing,

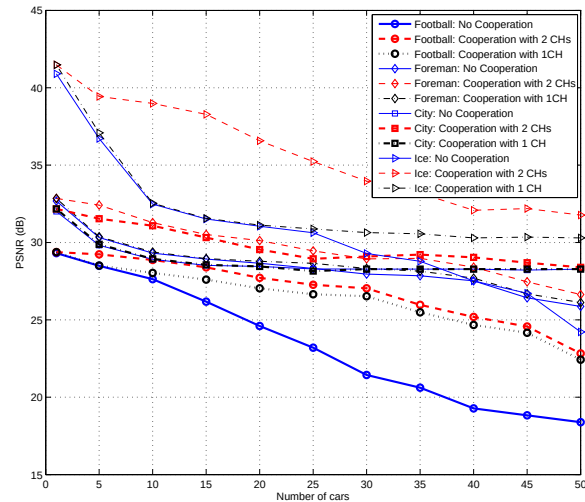


Fig. 5. PSNR vs. the number of vehicles.

with variations following a uniform distribution in the interval $[-1 + 1]$ km/hour at each TTI (without allowing them to go below 70 or above 100 km/hour). Vehicles move along these rules, with the clusters being formed dynamically. When the transmission of a GOP begins, any frames from the previous GOP that are not received by a vehicle are assumed lost, and the reception of the new GOP begins after performing error concealment as described in Section III. This process is repeated until the whole sequence is transmitted by the BS. Then, another iteration begins, and so on.

Section VI-A presents the V2V results, whereas the V2I results are presented in Section VI-B. In the V2I case, RSUs are placed along the side of the highway with a separation of 300 m between two consecutive RSUs, and a vehicle moving on the highway streams a video according to the methods of Section V. The vehicle and RSU transmit powers are set to 125 mW. The actual transmission range of the BS and RSUs depends on the received signal power at the vehicles after the signal travels the distance between the BS or RSU and the vehicle, and after being subjected to shadowing and fading. The association of vehicles to V2V clusters, or to LR LTE links with the BS or V2I links with RSUs, depends on the level of the received signal, or equivalently on the achievable communication rate, as governed by (12) and (13).

A. V2V Simulation Results

1) *Comparison of Different Video Sequences:* The QoS results, represented by PSNR performance, are presented in Fig. 5. The QoE results using the quality metric derived in (11), are shown in Fig. 6. In these figures, The horizontal axis represents the number of cars in the 1 km highway stretch. The PSNR and QoE metrics used in the vertical axis correspond to the average PSNR and QoE, averaged over all cars and iterations. Both figures show a remarkable superiority of the proposed collaborative approach compared to the non-collaborative scenario in terms of QoS and QoE, for the four video sequences. In the non-collaborative case, multicasting is performed over the LTE link by the BS to all vehicles all the

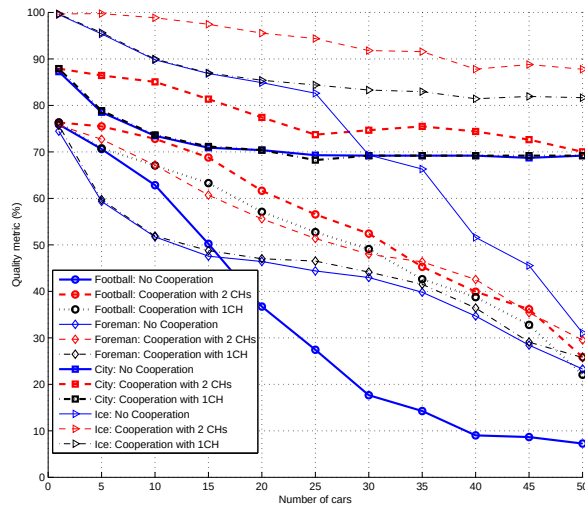


Fig. 6. Quality metric vs. the number of vehicles.

time, without resorting to V2V communications.

Comparing Foreman to Football, Fig. 5 shows that PSNR is generally higher for Foreman than Football. This is due to the relatively high motion in Football, which makes the sequence more sensitive to losses over the wireless channel, since the effect of error propagation becomes more serious. However, in terms of subjective video quality, estimated using the objective metric of (11), the QoE performance is generally better for Football in the collaborative case, until the number of users increases and the performance degrades due to the limitation of the multicasting rate by the user having the worst channel conditions.

This difference between PSNR and QoE performance can be interpreted as follows: When the base layer is received correctly in Football, then the bulk of the motion is captured. The received parts of the enhancement layer lead to providing a better quality. Hence, the user experience is more or less satisfactory while streaming the video, since the attention is generally focused on the fast moving part of the picture and mild errors in the rest of the picture can pass by unnoticed. With the Foreman sequence, user attention is focused on the whole image due to the slow motion in the sequence. Hence, losses that affect the visual quality of the image are noticed by the users who feel a negative impact, although PSNR performance may be better than Football. When the multicasting rate is significantly reduced due the increasing number of vehicles, losses start affecting the base layer in Football, which leads to a drastic performance degradation, as can be seen on the plot of the non collaborative case in Figs. 5 and 6.

This interpretation can also shed light on the performance of the partially collaborative approach. In fact, this partially collaborative approach where the base layer is sent to the CHs and the enhancement layer is multicast by the LTE BS leads to an intermediate performance between the fully collaborative (2 CHs per cluster) and the non collaborative scenarios. The performance of the partially collaborative approach is relatively close to the fully collaborative case with 2 CHs

per cluster for the Football sequence. However, it leads to a performance comparable to the non-collaborative scenario for the Foreman sequence. This behavior is noticed for both the QoS (Fig. 5) and QoE (Fig. 6) results. This can be explained as follows: With Football, this approach is sufficient to receive at least the base layer correctly through high data rates achievable at SR distances, which leads to the large gap with the non collaborative scenario, where losses occur in the base layer due to reduced LR multicasting rates when the number of vehicles increases. With Foreman, due to the low mobility, the base layer can be received correctly most of the time in both the collaborative (whether with one or two CHs) and non collaborative scenarios. The difference in performance with Foreman is measured by the ability to transmit more frames from the enhancement layer. This leads to a better performance for the collaborative scenario with 2 CHs where L_1 is transmitted on the SR, compared to the other scenarios where L_1 is multicast on the LR to all vehicles.

The results of City and Ice confirm the analysis presented for Football and Foreman in Figs. 5 and 6. City is a low mobility sequence, and its plots behave similarly to those of Foreman, in terms of relative performance between the cooperative scenario with 2 CHs, partial cooperation with 1 CH, and the non cooperative scenario. Ice contains higher motion, and thus the relative performance of the three scenarios (cooperative with 2 CHs, cooperative with 1 CH, and non cooperative) follows a trend similar to that of Football. In addition, since Football contains faster motion than Ice, it can be seen that the 1 CH scenario in the case of Ice performs similarly to the non cooperative scenario until the number of users increases and the performance of the non cooperative scenario decreases fast. In the case of Football, the 1 CH scenario has a superior performance even for low number of users, whereas for low motion sequences (Foreman and City), the performance between the 1 CH and the non cooperative scenarios is comparable for all the values investigated for the number of cars. It should be noted that the differences in the initial QoE and QoS levels (when there are no significant frame losses, e.g. when the number of cars is equal to one) of the different sequences are due to video source coding, where the coding of some sequences might lead to a better quality of the compressed sequence than the others. Source coding techniques are beyond the scope of this paper, where we used the JSVM encoder on the four sequences downloaded from: <ftp://ftp.tnt.uni-hannover.de/pub/svc/testsequences/>.

2) *Practical Implementation Issues*: It should be noted that, as mentioned in Section IV-A, it could be unnecessary to form the cluster when the traffic flow density is relatively low. Indeed, the proposed method takes this scenario into account. Moreover, in the simulations, the increase of the number of vehicles from one to 50 corresponds to a gradual increase in traffic density. In fact, in the limit scenario with one vehicle, there is obviously no clustering, and the plots for the collaborative and non-collaborative scenarios have the same result. When the number of cars increases, clustering is performed whenever it leads to gains, and not performed in case it is not beneficial. The results presented in the paper are averaged over many iterations in order to show the overall per-

formance of the compared methods. Consequently, when more and more cars collaborate, the chances of clustering increase, which explains the increasing performance gap between the cooperative approach with 2 CHs and the non-collaborative case, as the number of cars increases.

In addition, although the QoE and QoS results decrease with all methods, the performance gap (measuring the enhancement obtained) between the cooperative approach with 2 CHs and the non-collaborative case increases with the number of cars. However, the QoE and PSNR metrics themselves (not their difference with their no-coop counterparts) decrease for all methods when the number of cars increases. This is due to multicasting. In fact, to guarantee that all vehicles receive the data, multicasting is performed at the data rate achievable by the worst case user. This would be the worst case vehicle in a given cluster in the cooperative case, or the worst case car interested in receiving the video within the coverage area of the LTE BS in the non-collaborative case. Hence, when the number of cars increases, the probability of finding at least one with bad channel conditions increases, which leads to a performance degradation. Using unicasting, where the transmissions to each vehicle are independent from the transmissions to other vehicles (similarly to the case where the users are interested in different content, not the same video) cannot be a solution to the multicasting problem. Indeed, with unicasting, orthogonal channels are needed to avoid collisions over 802.11p or interference over LTE. With only four channels in 802.11p, it is clear that high quality real time video streaming simultaneously to a large number of requesting vehicles becomes very difficult. In addition, this will drain the wireless resources of the LTE BS in the non-cooperative case, which will significantly limit its capability to maintain other services or serve other users while distributing the SVC video.

Moreover, as explained in Section IV-A, when certain cars are very close to the BS, the performance of directly using the multicasting technique over LTE link could be better than that of using cooperative cluster method. Hence, the proposed method takes this scenario into account and would lead to $p_j = 0$ in (12). The simulation results presented in Figs. 5-8 are averaged over many iterations in order to show the overall performance of the compared methods: vehicles are moving on a highway, and during their trajectory they may pass next to a BS, then gradually move away, while the cluster formation and CH selection are being performed dynamically.

Finally, we note that statistics of packet loss might be misleading in terms of their relation to video PSNR and QoE. For example, in a GOP of 15 frames (one I-Frame and 14 P-Frames), losing a packet from Frame 1 and losing a packet from Frame 15 will give the same results in terms of packet loss statistics (one packet is lost per GOP in both cases). However, the loss of Frame 1 will lead to losing the whole GOP and thus will have significant implications on PSNR and QoE, whereas the loss of Frame 15 will have limited impact, since it will be replaced by Frame 14 and a new GOP will begin at Frame 16 with a new I-Frame. A thorough investigation is needed in order to analytically model the impact of packet losses on PSNR and/or QoE. In [22], an approach

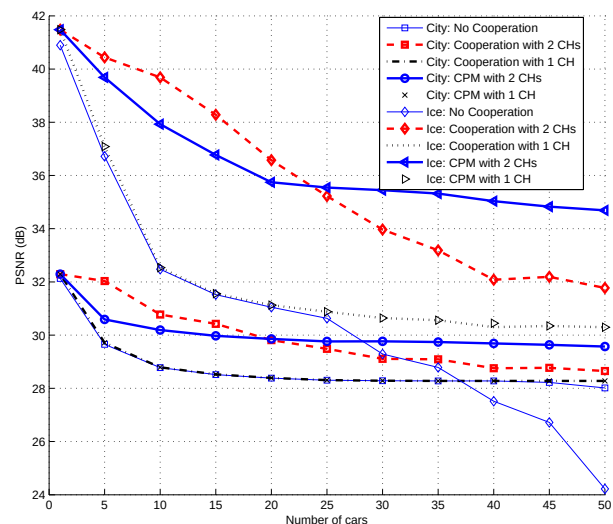


Fig. 7. Comparison with the CPM method: PSNR vs. the number of vehicles.

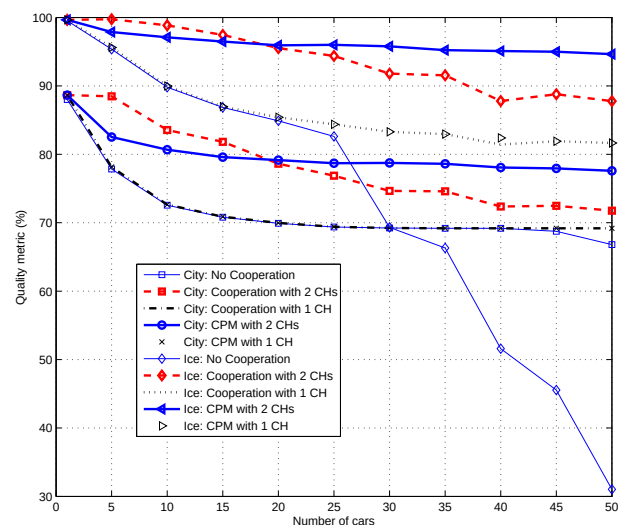


Fig. 8. Comparison with the CPM method: Quality metric vs. the number of vehicles.

is proposed to mathematically link the packet losses over the wireless channels to video loss distortion. This approach is presented for a single layer video with a single wireless link. Extending it to a multi-layer SVC video, in the presence of multihop transmissions using multiple technologies (LTE and 802.11), while performing dynamic clustering and resource allocation on these technologies in high speed and mobility environments is a very challenging task. It is indeed a very interesting topic for future research that is beyond the scope of this paper.

3) *Comparison to Other Clustering Methods:* The proposed approach is compared to a recent multi-CH method proposed in [9]. The approach of [9] consists of determining a single CH based on the position and speed information (Equations (1)-(5) [9]). It is denoted by CPM (Center Position and Mobility) and it assumes that all neighboring vehicles exchange this

information between each other prior to cluster formation. An extension to a multi-CH case is then proposed by subdividing the area into several subareas of equal dimensions, where a new CH is selected for each subarea. Consequently, the cars in each subarea are communicating with a single CH. Hence, even in the multi-CH case, the approach of [9] is not directly applicable to SVC video transmission, since two CHs are needed in each subarea in order to transmit two video layers. Therefore, we extend the approach of [9] as follows: (i) We subdivide the highway stretch into different subareas; (ii) We implement the single CH approach of [9] in each subarea to determine the primary CH in that subarea. This CH will be the one having the lowest “RPM” (Relative Position and Mobility) metric defined in Equation (5) of [9]; (iii) We select the car having the second best RPM metric as secondary CH to transmit the enhancement layer in the subarea. The comparison of this extended CPM method with the proposed approach is shown in Figs. 7 and 8 for the PSNR and QoE, respectively, using the City and Ice video sequences.

It can be seen that in the partial cooperation case with 1 CH, both methods perform comparably, and the results are nearly overlapping. However, in the cooperative scenario with 2 CHs, the proposed approach performs better than CPM when the number of cars is below 20-25. Above this range, the CPM method performs better. This is due to the fact that the CPM approach considers certain assumptions that are not included in the proposed approach. For example, the exchange of speed and position information would be generally done at the application layer. This would lead to defining smaller clusters with shorter transmission ranges on the average. Besides, in the simulations, we assume that is information is available all the time and that the CPM clusters can be updated on very fast time scales. Conversely to other methods in the literature (e.g. [9]), the proposed approach does not involve the exchange of position and speed information between the moving vehicles. Instead, cluster formation is performed based on physical layer (PHY) measurements. This can lead to faster and more dynamic cluster formations in high speed and mobility environments. In fact, in the LTE standard, feedback of channel state information can be sent to the BS at a granularity of 1 ms [14, 15], which allows the BS to determine the achievable LR data rates. With SR IEEE 802.11p communications, a car needs to detect the presence of a neighboring car using 802.11p, and determine the communication rate that can be achieved on their wireless link. This can be performed by PHY beacon or pilot transmissions, such that each car can determine the neighbors with which it can communicate, in addition to the achievable data rates. This does not require the exchange of position information, and could lead to “irregular” cluster formations governed by channel conditions rather than actual proximity. In fact, with this approach, a certain car A can be closer to car B than car C, but still achieve a better communication rate with car C than car B, due to fading fluctuations that might favor a link over another. This is confirmed by the example shown in Fig. 9.

Fig. 9 shows a snapshot for cluster formation and the association of cars to CHs. The example considered corresponds to 35 cars, placed initially on a highway stretch

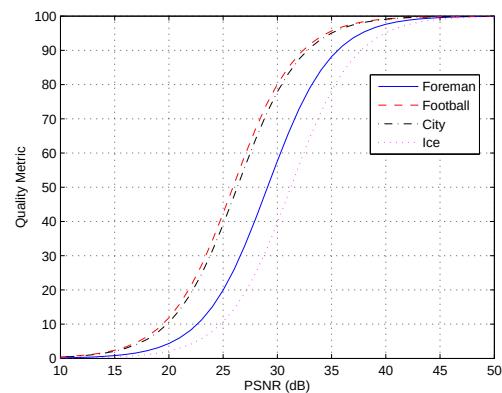


Fig. 10. Quality Metric Comparison.

TABLE III
PSNR (dB) AND QUALITY METRIC (%) RESULTS

	RSUs only	BS only	1 BS - 1 RSU
Foreman - PSNR	28.91	32.82	32.98
Football - PSNR	27.92	29.41	29.47
Foreman - Q_m	48.38	75.60	76.73
Football - Q_m	66.44	76.50	76.86

with horizontal coordinates between $[-500 + 500]$ m, the BS being positioned at (0,0). The cars move as described in the simulation model of Section VI, in the direction of increasing horizontal coordinates (It should be noted that to ensure clearer visibility of the figure, the axes dimensions are not at the same scale in Fig. 9). It can be seen that the cars are positioned between $[-300 + 800]$ m when the snapshot was taken. The top figure shows that with the proposed approach, one CH is receiving the base layer of the City sequence from the BS and multicasting it to all other cars. The enhancement layer is received by two CHs from the BS, and multicast by these two CHs to the other cars. The CH of the base layer is receiving the enhancement layer from the secondary CH positioned at (-62.1, 27.3) m. The bottom figure clearly shows the impact of the subdivision of the highway stretch into areas of length 400 m (200 m radius around CHs), with each area having two CHs: one to receive the base layer and another to receive the enhancement layer from the BS. Then, these CHs multicast their received data over V2V links to the other cars.

It can be seen that the CHs selected are not necessarily the closest cars to the BS. With the CPM method, this is expected since the selection is based on position and speed with respect to the other cars, independently from the distance to the BS. However, with the proposed approach, the selection takes into account channel fluctuations (fading and shadowing), which might favor a car located further away but having favorable fading conditions. Indeed, as future work, it would be interesting to investigate a combined approach where application layer information (speed and position) as in the CPM method can be exchanged on a slow time scale whereas fast channel information (PHY layer) can be used to tune the clusters and perform CH selection at a fast time scale (as in the proposed approach).

B. V2I Simulation Results

With V2I, we consider that each vehicle is moving independently (without communication with other vehicles). It

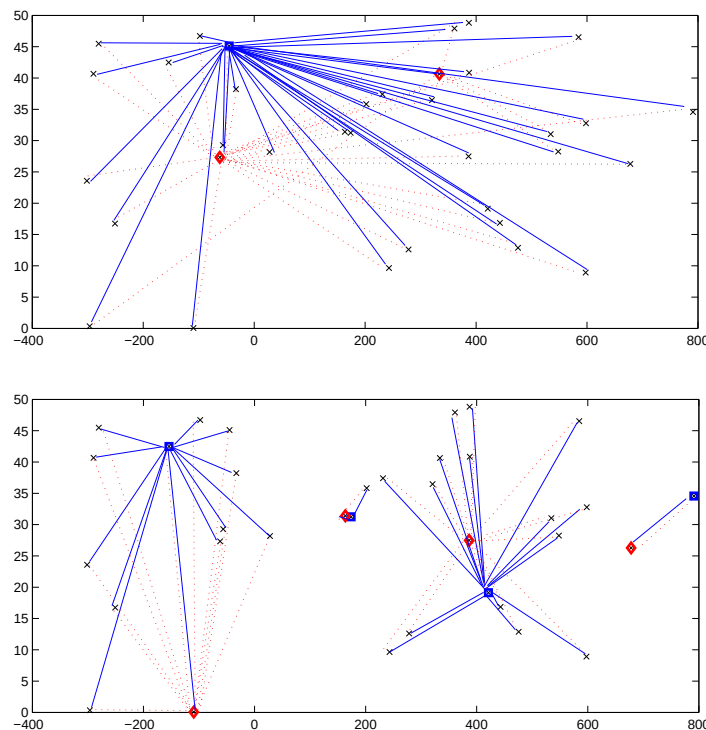


Fig. 9. Snapshot of Cluster formation and CH selection with 35 cars and the City sequence. Top: Proposed approach. Bottom: CPM method [9]. Blue squares represent the primary CHs (best set of CHs), the red diamonds the secondary CHs (second best set), and 'x's represent cluster members. Solid lines represent connections between cluster members and primary CHs, whereas dotted lines represent the connections between cluster members and secondary CHs. The horizontal axis corresponds to the length of the highway stretch, and the vertical axis to the highway width. The BS is assumed to be at position (0,0).

attempts to download the video for real time streaming using one of the methods described in Section V.

The obtained results for the Foreman and Football sequences are shown in Table III. For both sequences, the best performance is achieved by Scenario 3, where the best LR (LTE RB) and SR (RSU) resources are selected to transmit the two layers. Comparing the two sequences, Table III shows that Foreman has better QoS (PSNR) performance, since Football contains higher motion and hence errors from frame losses propagate more seriously. However, QoE (Q_m) results are better for Football. These are justified by the interpretations presented in Section VI-A. Furthermore, they can be mathematically confirmed using the results of [5], where different sets of optimal parameters (b_1, b_2) were obtained during QoE investigation for each video sequence. Thus the plot of (11), shown in Fig. 10, indicates better user experience for Football compared to Foreman, when the same PSNR is achieved.

In all cases, the best V2I performance was obtained by joint collaboration between LTE BS and 802.11p RSU. This differs from the V2V results (where the best results are obtained with 2 CHs) since the RSUs are fixed. Vehicles moving dynamically can have vehicular CHs in their clusters at closer proximity than RSUs, which leads to higher achievable multicasting rates on the SR, thus more frames transmitted successfully and consequently better QoS/QoE performance.

VII. CONCLUSIONS

Collaborative SVC video streaming methods in vehicular networks were proposed using V2V communications. The proposed techniques are based on grouping vehicles into collaborative clusters, and on varying the cluster head dynamically as the wireless channel conditions change. LTE was used to transmit the video to cluster heads and IEEE 802.11p was used for the communication between vehicles. The collaborative techniques were extended to the V2I communication scenario. Simulation results with four different video sequences with varying characteristics were presented and analyzed. The proposed methods were shown to lead to significant enhancements in PSNR and QoE metrics compared to the traditional non collaborative scenarios. In addition, their performance tradeoffs compared to recent methods from the literature were discussed.

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